

Aloysius Allen Darmawan

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EDUCATION

Monash University

Feb. 2022 – Jul. 2026

Bachelor of Science (Pure Mathematics, Physics)/Bachelor of Computer Science

Melbourne, VIC

- Weighted Average Mark (WAM): Distinction average

EXPERIENCE

Software Developer Intern

Jun. 2025 - Jan. 2026

Geotesta

Melbourne, VIC

- Proposed and prototyped a PyTorch ResNet classifier for tunnel face rock structure, liaising with geotechnical engineers to define Geological Strength Index-based classes; reached roughly 80% test accuracy
- Developed a Flask/Bootstrap tailings dam analysis module with an integrated LLM chatbot for ConVue, Geotesta's cloud monitoring platform used by engineers across multiple firms

Backend Engineer Intern

Nov. 2024 – Feb. 2025

ANZ Plus

Melbourne, VIC

- Streamlined CI/CD pipelines by building GitHub Actions workflows for release tagging and deployment validation
- Developed and deployed Go cloud-native microservices for real-time credit decisioning, automating SQL generation to query BigQuery datasets with 100M+ records resulting in scalable, data-driven lending decisions

Student Software Engineer

Aug. 2023 – Feb. 2025

Monash Connected Autonomous Vehicle

Melbourne, VIC

- Implemented ROS 2 launch systems in Python and C++ to synchronise 8 LiDAR sensors on a Nissan e-NV200 van, achieving deterministic and real-time sensor fusion for robust autonomy of the van
- Collaborated with multidisciplinary subteams and stakeholders to integrate autonomous valet parking capabilities across the Autoware (ROS 2) stack, coordinating technical requirements and system interfaces

COMMUNITY AND LEADERSHIP

Mental Health Immersive Technologies Developer

GitHub | Feb. 2026 – Apr. 2026

EUGLOH

Paris, France

- While on a study abroad semester at Lund University in Sweden, was competitively selected for a program about bridging art therapy and mental health using immersive technologies at the University of Paris-Saclay in France
- Integrated TouchDesigner modules, an Arduino microcontroller, and scripts in Python and C++ to develop a Spatial Augmented Reality (SAR) environment for people suffering from chronic pain

University Open Day Student Ambassador

Aug. 2024 – Aug. 2025

Monash University

Melbourne, VIC

- Represented the Faculty of Information Technology as a student ambassador at outreach events such as Open Day, guiding 300+ prospective students and their families through degree offerings, job prospects and university life

PROJECTS

All about the Sky | *Next.js, React, TypeScript, Tailwind CSS, HTML, Open-Meteo API*

Demo | **GitHub**

- Built a weather-driven animated sky with React, TypeScript, and the Canvas 2D API, mapping live Open-Meteo data to procedural clouds, rain, and day/night cycles, with mouse and touch interaction via Pointer Events
- Reduced upstream weather API calls to at most 4 per hour per 11 km area by building a cached Next.js route that proxies Open-Meteo, validates inputs, rounds coordinates for privacy, and caches for 15 minutes

Avian Annotator | *React, Spring Boot, Node.js, PostgreSQL, AWS S3, Docker*

GitHub

- Oversaw the backend development of an accessible web-based bird image annotation platform allowing ornithology researchers and citizen scientists to collaboratively create high-quality training datasets for AI models
- Designed and deployed RESTful APIs using Spring Boot and PostgreSQL to efficiently query and manage multi-user annotation data, while integrating AWS S3 for secure, scalable image storage

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks/Libraries: React, Node.js, Spring Boot, Flask, PyTorch, Bootstrap, ROS 2, Tailwind CSS

Mathematics/Physics: Linear Algebra, Probability, Statistics, Graph Theory, Calculus, Fluid Mechanics